



SYSTEMS INTEGRATION & LOW-VOLTAGE ENGINEERING

AZKA SMART

Intelligent Living & Systems Engineering for the Middle East

CERTIFIED: KNX TP1 • ZIGBEE 3.0 • MATTER OVER THREAD • MODBUS RTU • DALI-2

Azka Smart is a specialized smart-home engineering and AI technology firm operating across Cairo, Riyadh, and Dubai. We design, build, and commission industrial-grade residential automation: hybrid wired/wireless bus topologies, native VRF chiller gateways, enterprise VLAN network segregation, and proprietary AI-driven CAD floor plan analysis.

[KNX Partner]

[CSA Member]

[DALI-2]

[CEDIA Member]

100%

LOCAL OFFLINE EXECUTION

<30ms

BUS SWITCH LATENCY

55°C

AMBIENT THERMAL TOLERANCE

35%+

AVERAGE ENERGY REDUCTION

REGIONAL BUREAUS: Cairo (Tower 4, 90th St) • Riyadh (King Fahd Rd) • Dubai (Marina Plaza L24)

DOCUMENT REF: AS-ENG-SPEC-2026-V4 | SECURITY: PUBLIC

www.azkasmart.com

SECTION 01: SYSTEM ARCHITECTURE & TOPOLOGY

Four-Tier Industrial Fieldbus Topology

Designed to eliminate single points of failure with 100% local survivability during internet outages.

TIER 01: EDGE & LOCAL PROCESSING LAYER

Industrial Edge Core (Dual Cortex-A72, 4GB LPDDR4, Local SQLite/Time-Series DB)

Zero cloud dependency for daily routines. All scenes, logic scripts, thermostat PID loops, and inter-device triggers run 100% locally on dedicated hardware in the home distribution board.

- Survivability: 100% functional during ISP outages
- Switch Response: < 25 milliseconds
- Data Privacy: Local encrypted storage

TIER 02: FIELDBUS & WIRELESS MESH LAYER

KNX TP1 (9.6kbps shielded twisted pair) • Zigbee 3.0 (IEEE 802.15.4) • Matter over Thread

Dual-path topology: Hardwired KNX twisted pair for foundational lighting/HVAC lines, combined with 2.4GHz Zigbee 3.0 / Thread mesh for retrofits and auxiliary sensors with self-healing routing.

- Topology: Star, Line, Tree (KNX) + Self-healing Mesh
- Max Hop Count: 30+ nodes per Zigbee coordinator
- Interference Resilience: Adaptive Frequency Hopping

TIER 03: HVAC & THERMAL ENGINEERING LAYER

Modbus RTU RS-485 / BACnet IP / CoolMaster VRV/VRF Gateways

Native protocol integration with commercial multi-split and VRF systems (Daikin, Mitsubishi, LG, Carrier, Gree). Bidirectional control over compressor speed, fan step, error codes, and dew-point threshold.

- Control Depth: Setpoint, Fan Speed, Louver, Error Diagnostics
- Thermal Range: Engineered for 50°C+ summer ambients
- Energy Optimization: Adaptive scheduling based on occupancy and weather

TIER 04: POWER & PANEL INFRASTRUCTURE LAYER

DIN-Rail Modular Actuators (ABB/Schneider/Hager) • Zero-Cross Switching Relays

Engineered for high inrush capacitive currents from modern architectural LED drivers (up to 80A peak). Type 2 40kA surge arresters and dedicated neutral topology compliant with regional electric codes.

- Relay Rating: 16A continuous / 80A 20ms inrush (Zero-Cross)
- Surge Protection: Type 2 SPD 40kA (8/20µs) with telemetry
- Grounding: Dedicated equipotential bonding system

Note: Complete single-line wiring diagrams (SLD) and riser schematics provided upon site CAD ingestion.

SECTION 02: REGIONAL ELECTRICAL STANDARDS & COMPLIANCE

Middle Eastern Power Grid Engineering Matrix

Comprehensive technical accommodation for grid frequency variations, neutral wiring, and ambient thermals.

EGYPT

GRID: 220V AC ± 10%, 50 Hz

STANDARDS: EEHC & Egyptian Building Code

NEUTRAL CONDUCTOR STRATEGY:

Neutral wire retrofits or bypass capacitance on legacy live-only backboxes

THERMAL & ENVIRONMENTAL SAFEGUARDS:

High summer dust resistance (IP54+ on perimeter sensors), dust-proof switch membranes

SAUDI ARABIA

GRID: 230V AC ± 10%, 60 Hz (3-Phase 400V)

STANDARDS: SASO Compliant & SEC Technical Guidelines

NEUTRAL CONDUCTOR STRATEGY:

Compulsory neutral conductors at all switch backboxes, Type G BS 1363 sockets

THERMAL & ENVIRONMENTAL SAFEGUARDS:

Extreme 55°C heat rating, heavy-duty VRF Modbus zoning, Najdi thick-wall RF repeaters

UNITED ARAB EMIRATES

GRID: 230V AC ± 6%, 50 Hz

STANDARDS: DEWA / ADDC Green Building Regulations

NEUTRAL CONDUCTOR STRATEGY:

Low-smoke zero-halogen (LSZH) cabling, mandatory sub-metering telemetry

THERMAL & ENVIRONMENTAL SAFEGUARDS:

Continuous 24/7 humidity control to prevent condensation on chilled ceiling beams

SECTION 03: VERIFIED CASE STUDIES & FIELD DATA

Commissioned Technical Deployments

Empirical data from live residential installations in New Cairo, Riyadh, and Palm Jumeirah.

Palatial Residence - New Cairo

NEW CAIRO, EGYPT | Residential Villa (1,450 m² BUU - 3 Floors)

148 KNX DALI-2 dimming circuits, 14 VRF AC zones via Modbus RTU, 24 automated roller curtains, solar inverter RS485 telemetry, biometric facial entry, 100% local automation rack.

LIGHTING CIRCUITS

148 Channels

ENERGY REDUCTION

34.2%

CLOUD DEPENDENCY

0% (Local Bus)

SWITCH LATENCY

18 ms

Executive Villa - Al-Malqa District

RIYADH, SAUDI ARABIA | Contemporary Residence (880 m² BUU)

Full Matter-over-Thread and Zigbee 3.0 mesh with zero conduit demolition in finished Najdi stone masonry. DIN-rail multi-channel power meters with 60Hz grid filtering, automated smart irrigation tied to meteorological evapotranspiration sensors.

MESH NODES

86 Devices

WATER SAVING

42%

GRID FREQUENCY

60 Hz SASO

INSTALL TIME

6 Days (No Masonry Work)

Skyline Penthouse - Palm Jumeirah

DUBAI, UAE | High-Rise Penthouse (620 m²)

Dual-bus KNX & Lutron Palladiom keypad integration. 10-zone multi-room high-resolution audio over IP (Dante), electrochromic smart privacy glass synchronized with solar tracking, building management system (BMS) BACnet bridge.

AUDIO ZONES

10 Zones IP

HVAC INTEGRATION

DEWA / BACnet IP

SOLAR HEAT BLOCK

68% Solar Gain

CONTROL SURFACE

Solid Brass Keypads

SECTION 04: AI-ASSISTED DESIGN & CAD ENGINE

Automated DWG Blueprint Ingestion & BOQ

How Azka Smart automates electrical load estimation and line-item contractor quotes.

CAD & DWG Semantic Blueprint Parser

Vision-transformer model detects room volume, wall materials (reinforced concrete vs drywall vs masonry), and automatically calculates lux requirements and sensor blind spots.

Live Regional Supplier Grounding

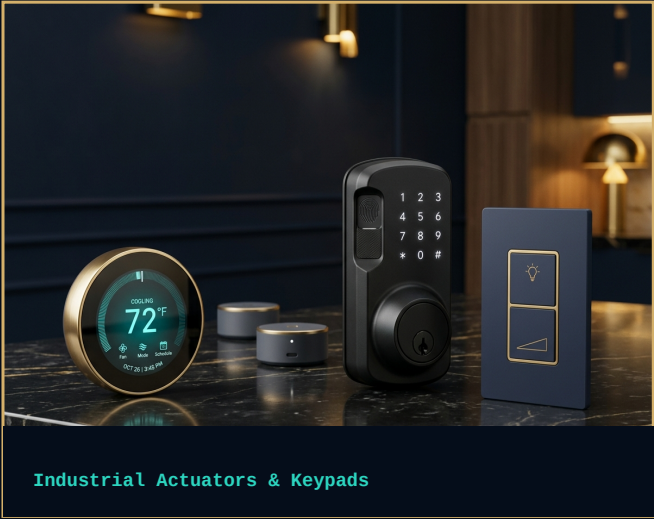
Integrated directly with official component catalogs in Cairo (EGP) and Riyadh (SAR), ensuring exact real-time component availability, warranty tiers, and contractor pricing.

Predictive Middle East HVAC Optimizer

Analyzes thermal building envelope mass and regional hourly weather forecasts to pre-cool zones before peak ambient temperatures hit, cutting compressor peak wear.

Edge Anomaly & Telemetry Monitor

Real-time edge telemetry monitoring electrical phase balance, neutral wire current leakage, and actuator relay operating cycles for predictive preventive maintenance.



SECTION 05: PHYSICAL BUREAUS & WORKSHOP FACILITIES

Direct Field Presence & Engineering Labs

Certified panel pre-wiring workshops and central distribution hubs in Egypt, KSA, and UAE.



Cairo, Egypt

ENGINEERING HQ, R&D & CENTRAL DISTRIBUTION

Address: Tower 4, Northern 90th St, New Cairo

Group headquarters hosting our core embedded software engineers, panel assembly workshop, QA testing lab, and 1,200m² central warehouse serving projects across Egypt and North Africa.

Email: cairo@azkasmart.com | Direct: +20 150 189 6456



Riyadh, Saudi Arabia

KSA OPERATIONS & VISION 2030 PROJECT DELIVERY

Address: King Fahd Road, Al-Sahafa District, Riyadh

Dedicated engineering field bureau delivering large-scale villa developments, private estates, and luxury compound projects fully compliant with Saudi Building Code (SBC) and SASO standards.

Email: riyadh@azkasmart.com | Direct: +966 50 000 0000



Dubai, United Arab Emirates

EXPERIENCE CENTER & HIGH-END INTEGRATIONS

Address: Marina Plaza Level 24, Dubai Marina, Dubai

Interactive live showroom demonstrating native DALI-2 architectural lighting scenes, multi-room audio acoustics, and BMS integration for high-rise residences and hospitality developments in the UAE.

Email: dubai@azkasmart.com | Direct: +971 4 000 0000

Submit Architectural CAD / DWG for Engineering Assessment

Email your project floor plans and electrical single-line requirements directly to our engineering bureau. Our AI pipeline will generate a preliminary load analysis, device schedule, and line-item contractor BOQ within 24 business hours.

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